

GAUTAM BUDDHA UNIVERSITY



SCHOOL OF ICT

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DETAILED SYLLABI (SEMESTER III-IV)

INTEGRATED B.TECH.-M.TECH. (COMPUTER SCIENCE AND ENGINEERING)

ACADEMIC BATCH: 2025-2030

AND

B.TECH. (COMPUTER SCIENCE AND ENGINEERING)

& SPECIALIZATIONS IN

DATA SCIENCE AND CYBER SECURITY

ACADEMIC BATCH: 2025-2029

35th BoS Date: 30-July-2026

SEMESTER III

Scientific Computing with Python			
Course Code:	CSE201	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Develop proficiency in Python for scientific and engineering applications.			
2. Introduce numerical methods and computational techniques for solving scientific problems.			
3. Apply scientific Python libraries for numerical computation, optimization, visualization, and data analysis.			
4. Develop efficient, reproducible, and scalable scientific computing solutions.			
5. Enable students to solve real-world engineering and data-driven computational problems.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Develop Python programs for scientific and engineering computations.			
2. Apply numerical methods using NumPy and SciPy to solve computational problems.			
3. Analyze, visualize, and interpret scientific datasets using Python libraries.			
4. Develop optimization and machine learning solutions for scientific applications.			
5. Design efficient, reproducible, and scalable scientific computing workflows using modern Python tools.			

UNIT I: FOUNDATIONS OF SCIENTIFIC COMPUTING AND PYTHON

Introduction to Scientific Computing, Python programming for scientific applications, Python data structures and control flow, Functions, modules, packages, exceptions, Object-Oriented Programming in Python, Jupyter Notebook environment, Scientific Python ecosystem, Introduction to version control using Git, Libraries: Python, Jupyter, IPython.

UNIT II: NUMERICAL COMPUTING WITH NUMPY AND SCIPY

NumPy arrays and vectorized programming, Matrix operations, Broadcasting, Random number generation, Linear algebra using NumPy, Sparse matrices, Numerical differentiation, Numerical integration, Root finding, Interpolation, Solving ordinary differential equations (ODEs), Libraries: NumPy, SciPy.

UNIT III: SCIENTIFIC DATA ANALYSIS AND VISUALIZATION

Scientific data formats, Data manipulation using Pandas, Data cleaning and pre-processing, Statistical analysis, Exploratory Data Analysis, Scientific visualization, 2D and 3D plotting, Heatmaps, Interactive visualization, Visualization best practices, Libraries: Pandas, Matplotlib, Plotly, Seaborn.

UNIT IV: OPTIMIZATION AND SCIENTIFIC MACHINE LEARNING

Mathematical optimization concepts, Linear and nonlinear optimization, Constrained optimization, Curve fitting, Least-squares estimation, Introduction to scientific machine learning, Feature engineering, Libraries: SciPy Optimize, Scikit-learn.

Signature

Signature

Signature

UNIT V: ADVANCED SCIENTIFIC COMPUTING

Graph and network analysis, Parallel computing concepts, Performance optimization, Profiling Python code, Introduction to GPU computing, Unit testing, Package development, Reproducible computational research, Applications of scientific computing involving simulation, optimization, data analysis, Libraries: NetworkX, multiprocessing, Overview of Numba/CuPy, pytest.

Text Books:

1. Hans Petter Langtangen, *A Primer on Scientific Programming with Python* (5th Edition), Springer International Publishing, 2016.
2. Jake VanderPlas, *Python Data Science Handbook: Essential Tools for Working with Data* (2nd Edition), O'Reilly Media, 2016.
3. Wes McKinney, *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython* (3rd Edition), O'Reilly Media, 2022.
4. Christian Hill, *Learning Scientific Programming with Python* (2nd Edition), Cambridge University Press, 2020.
5. Eric Matthes, *Python Crash Course: A Hands-On, Project-Based Introduction to Programming* (3rd Edition), No Starch Press, 2023.

Handwritten signatures:
Aulanki, Vandana, [unclear]
[unclear]

Data Structures			
Course Code:	CSE203	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Understand the fundamental concepts of data structures, abstract data types, recursion, and algorithm analysis using asymptotic notations.			
2. Analyze, implement, and compare searching, sorting, and hashing techniques based on their efficiency and suitability for different applications.			
3. Design and implement linear data structures such as linked lists, stacks, and queues, and apply them to solve computational problems.			
4. Apply tree-based data structures, including binary trees and binary search trees, to perform efficient storage, traversal, searching, and expression evaluation.			
5. Represent, traverse, and analyze graph data structures using standard graph traversal algorithms for solving connectivity and ordering problems.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Explain the fundamentals of data structures and algorithm analysis.			
2. Implement and analyze searching, sorting, and hashing techniques.			
3. Apply linked lists, stacks, and queues to solve computational problems.			
4. Construct and manipulate tree data structures for efficient data processing.			
5. Represent and traverse graphs using standard graph algorithms.			

UNIT I: INTRODUCTION TO DATA STRUCTURES

Introduction to Data Structures, Abstract Data Types, Linear and Non-linear Data Structures, Arrays as ADT, One Dimensional Array, Basic Operations on Arrays, Recursion: Introduction and Process, Examples: Factorial, Fibonacci Sequence, Tower of Hanoi; Asymptotic analysis introduction, Asymptotic notations, Big-Oh Notation, Time Complexity.

UNIT II: SEARCHING AND SORTING

Searching: Linear Search, Binary Search; Sorting: Introduction, Efficiency Considerations, Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort, Counting Sort, Radix Sort, Bucket Sort.

UNIT III: LINKED LIST, STACKS AND QUEUES

Linked List: Introduction, Singly Linked List, Basic Operations on Singly Linked List: Insertion, Deletion, Traversal, Searching, Circular Linked List, Basic Operations on Circular Linked List: Insertion, Deletion, Traversal, Introduction to Doubly Linked List. Stacks: Introduction, Insertion, Deletion, Traversal, Applications: Infix to Postfix Conversion, Post-fix Expression Evaluation, Queues: Introduction, Insertion, Deletion, Traversal, Circular Queue: Insertion, Deletion, Traversal.

UNIT IV: TREES

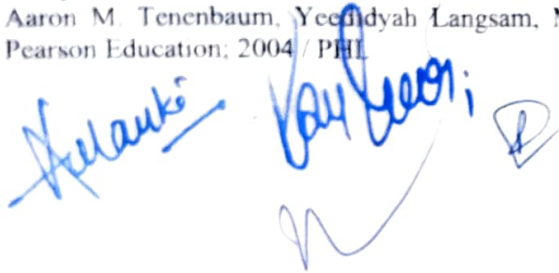
Tree Data Structures, Height, Complete, Full, Perfect Trees, Binary Trees, Properties of Binary Trees, Representation, Traversal: Pre-Order, In-Order, Post-Order, Binary Search Trees, BST Searching, BST Insertion, BST Deletion, Constructing Tree from given Traversals, Evaluating an expression tree, Heaps, Min-Max Heaps, Heap Tree, Heap Sort, Heap as Priority Queue, Insertion, Deletion.

UNIT V: HASHING AND GRAPHS

Hashing: Hash function, Hash Tables, Hashing Collision Resolution Strategies: Separate Chaining, Linear probing, Quadratic probing, Double hashing; Graphs, Introduction, Different Types of Graphs, Graphs Representations, Incidence Matrix, Adjacency Matrix, Graphs Traversals: Breadth First Search, Depth First Search, Applications: Connected Components, Topological Sort.

Text Books:

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, Introduction to algorithms. MIT Press; 1990 / McGraw-Hill.
2. E. Horowitz, S. Sahni, Fundamentals of data structures, Computer Science Press, 1982.
3. N. Karumanchi, Data Structures and Algorithms Made Easy, CareerMonk Publications, 2017.
4. S. Lipschutz, Data Structures With C, Tata McGraw-Hill, 2011.
5. Aaron M. Tenenbaum, Yeeendyah Langsam, Moshe J. Augenstein, Data Structures Using C, Pearson Education; 2004 / PHI.



Database Management System			
Course Code:	CSE205	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3

COURSE OBJECTIVES

1. Describe the fundamental elements of relational database management systems
2. Explain the basic concepts of relational data model, entity-relationship model, relational database design, relational algebra and SQL.
3. Design ER-models to represent simple database application scenarios
4. Convert the ER-model to relational tables, populate relational database and formulate SQL queries on data.
5. Improve the database design by normalization.

COURSE OUTCOMES

At the end of the course the students should be able to:

1. Students will have an understanding of database concepts and thorough knowledge of database software's.
2. Students will be able to model an application's data requirements using ER diagrams
3. Students will be able to write SQL commands to create tables and query data in a relational DBMS
4. Students will be able to execute various advanced SQL queries related to transactions, concurrency
5. Students will be able to explain the principle of transaction management design.

UNIT I: DATABASE SYSTEM

Data base system vs. file system, view of data, data abstraction, instances and schemas, data models, ER model, relational model, DDL, DML, database access for applications programs, data base users and administrator, transaction management, data base system structure, storage manager, query processor, history of data base systems, data base design and ER diagrams, beyond ER design entities, attributes and entity sets, relationships and relationship sets, additional features of ER model, concept design with the ER model, and conceptual design for large enterprises.

UNIT II: RELATIONAL MODEL

Introduction to the relational model, integrity constraint over relations, enforcing integrity constraints, querying relational data, and logical data base design, destroying /altering tables and views. relational algebra and calculus: relational algebra, selection and projection set operations, renaming, joins, division.

UNIT III: BASIC SQL QUERY

Examples of basic SQL queries, nested queries, correlated nested queries set, comparison operators, aggregative operators, NULL values, comparison using null values, logical connectivity's, AND, OR and NOT, impact on SQL constructs, outer joins, disallowing NULL values, complex integrity constraints in SQL triggers and active data bases.

UNIT IV: SCHEMA REFINEMENT

Problems caused by redundancy, decompositions, problem related to decomposition, reasoning about FDS, FIRST, SECOND, THIRD normal form, BCNF, forth normal form, lossless join decomposition.

dependency preserving decomposition, schema refinement in data base design, multi valued dependencies.

UNIT V: OVERVIEW OF TRANSACTION MANAGEMENT

ACID properties, transactions and schedules, concurrent execution of transaction, lock based concurrency control, performance locking, and transaction support in SQL, crash recovery, concurrency control, Serializability and recoverability, lock management, lock conversions, dealing with deadlocks, specialized locking techniques, concurrency without locking, crash recovery: ARIES, log, other recovery related structures, the write, ahead log protocol, check pointing, recovering from a system crash, media recovery, other approaches and interaction with concurrency control.

Text Books:

1. Elmasri Navrate, Data Base Management System, Pearson Education, 2008.
2. Raghurama Krishnan, Johannes Gehrke, Data Base Management Systems, TMH, 3rd edition, 2008.
3. C. J. Date, Introduction to Database Systems, Pearson Education, 2009.
4. Silberschatz, Korth, Database System Concepts, McGraw hill, 5th edition, 2005.
5. Rob, Coronel & Thomson, Database Systems Design: Implementation and Management.

Computer Organization and Architecture			
Course Code:	CSE207	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To understand the basic organization of computers, data representation, and arithmetic operations.			
2. To study machine instructions, addressing modes and instruction set architectures.			
3. To learn processor organization and control unit design techniques.			
4. To understand memory organization, cache memory, and virtual memory management.			
5. To understand I/O organization, data transfer methods, and peripheral interfaces.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Understand the basic organization and functional units of a computer system, including data representation, arithmetic operations, and ALU organization.			
2. Explain machine instructions, addressing modes, instruction formats, and compare RISC and CISC architectures			
3. Analyze processor organization, instruction execution cycle, and the design of hardwired and micro-programmed control units.			
4. Understand memory organization, cache memory, semiconductor memories, and virtual memory management techniques, the concept of pipelining and pipeline hazards			
5. Describe I/O organization, data transfer mechanisms, I/O interfaces, and the functioning of peripheral and secondary storage devices.			

UNIT I: Basic Computer Organization and Data Representation

Functional units of a computer, Basic computer organization and block diagram, Bus structures and interconnections, Register transfer language (RTL), Register transfer and micro-operations, Number systems and base conversions, Signed number representation, Fixed-point and floating-point representation, Arithmetic algorithms, Arithmetic Logic Unit (ALU) organization.

UNIT II: Instruction Set and Addressing Modes

Representation of Instructions Representation of Instructions: Machine instruction Operands, Addressing: Machine instructions, Operands, Addressing modes, Instruction formats, Instruction sets, Instruction set architectures - CISC and RISC architectures

UNIT III: Processor Organization and Control Unit Design

Processing Unit: Organization of a processor - Registers, ALU and Control unit, Data path in a CPU, Instruction cycle, Organization of a control unit - Operations of a control unit, Hardwired control unit, Micro-programmed control unit.

UNIT IV: Memory Organization and Pipelining

Memory Subsystem: Semiconductor memories, Memory cells - SRAM and DRAM cells, Internal Organization of a memory chip, Memory organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory, Memory Management Hardware, Instruction pipelining, pipeline hazards.

UNIT V: Input/Output Organization and Peripheral Devices

Input/Output Subsystem: Access of I/O devices, I/O ports, I/O control mechanisms - Program controlled I/O Interrupt controlled I/O and DMA controlled I/O I/O interfaces Program controlled I/O, Interrupt controlled I/O, and DMA controlled I/O, I/O interfaces - Serial port, Parallel port, PCI bus, SCSI bus, USB bus, I/O peripherals - Input devices, Output devices, Secondary storage devices.

Text/Reference Books:

1. Computer Organization and Design: The Hardware/Software Interface, Authors: David A. Patterson and John L. Hennessy Publisher: Morgan Kaufmann Edition: 2014
2. Structured Computer Organization, Author: Andrew S. Tanenbaum, Todd Austin Publisher: PHI Learning, Edition: 2016
3. Computer System Architecture, Author: M. Morris Mano, Publisher: Prentice Hall of India Edition: Third Edition, 2002
4. Computer Organization and Architecture: Designing for Performance, Author: William Stallings Publisher: Pearson Education, Edition: 2022
5. Computer Organization, Authors: Carl Hamacher, Zvonko Vranesic and Safwat Zaky, Publisher: McGraw Hill, Edition: 2001

Operating System			
Course Code:	CSE209	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Understanding how Operating System is Important for Computer System.			
2. To make aware of different types of Operating System and their services.			
3. To learn different process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.			
4. To know virtual memory concepts and secondary memory management.			
5. Understanding of Security & protection in Operating System.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Understands the different services provided by Operating System at different level			
2. They learn real life applications of Operating System in every field.			
3. Understands the use of different process scheduling algorithm and synchronization techniques to avoid deadlock.			
4. They will learn different memory management techniques like paging, segmentation and demand paging etc.			
5. Perform implementation of protection mechanisms in operating system			

UNIT I: INTRODUCTION TO OPERATING SYSTEM

Importance of operating systems, basic concepts and terminology about operating system, memory management, processor management, device management, information management functions.

UNIT II: PROCESS MANAGEMENT

Elementary concept of process, job scheduler, process scheduling, operation on process, threads, overview, scheduling criteria, scheduling algorithms, algorithm, and deadlocks: system model, deadlock characterization, deadlocks prevention, deadlocks avoidance, deadlocks detection, recovery from deadlock.

UNIT III: MEMORY MANAGEMENT

Memory management, logical versus physical address space, swapping, contiguous allocation, paging, segmentation, demand paging, page replacement, page replacement algorithms, allocation of frames, thrashing, demand segmentation.

UNIT IV: STORAGE MANAGEMENT

File concept, directory structure, protection, file-system structure, allocation method, free-space management, directory implementation.

UNIT V: I/O SYSTEMS

I/O hardware, Application of I/O interface, Overview of Kernel I/O subsystem, three types of I/O systems, memory based I/O, I/O based I/O, peripheral based I/O.

Text Books:

- Galvin, Wiley, Operating Systems Concepts, 8th edition, 2009.
- James L Peterson, Operating Systems Concept, John Wiley & Sons Inc, the 6Rev edition, 2007.

Handwritten signature

36th BoS Date: 30-July-2026

Academic Batch: 2025 and Onwards

Reference Books:

1. Deitel H. M., An Introduction to Operating Systems, Addison-Wesley, 1990.
2. Stallings William, Operating Systems, PHI, New Delhi, 1997.
3. S. Tanenbaum Modern Operating Systems, Pearson Education, 3rd edition, 2007.
4. Nutt, Operating System, Pearson Education, 2009.
5. S. Tanenbaum, Distributed Operating Systems, Prentice Hall, 2nd edition, 2007.

Arunkumar
R
W

Scientific Computing with Python Lab			
Course Code:	CSE281	Course Credits:	2
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Labs (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Labs:	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Develop proficiency in using Python and scientific computing libraries for solving computational problems.			
2. Apply numerical methods and scientific programming techniques to analyze and solve engineering and scientific problems.			
3. Perform data processing, visualization, and optimization using modern Python tools and libraries.			
4. Design, implement, and evaluate scientific computing solutions through practical experimentation and real-world applications.			
5. Develop analytical, programming, and problem-solving skills for scientific and engineering computations.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Develop Python programs using scientific computing libraries such as NumPy, SciPy, Pandas, and Matplotlib for computational tasks.			
2. Apply numerical methods and matrix computations to solve scientific and engineering problems.			
3. Perform data analysis, visualization, and optimization using appropriate Python libraries.			
4. Design, implement, and evaluate scientific computing solutions for real-world engineering and data-driven applications.			
5. Analyze computational performance and develop efficient, reliable, and reproducible scientific computing programs.			

LIST OF EXPERIMENTS:

1. Write a Python program to perform basic scientific computations using variables, operators, functions, and control statements in Jupyter Notebook.
2. Write a Python program to create and manipulate NumPy arrays, perform vectorized operations, and compare their performance with Python lists.
3. Implement matrix operations (addition, multiplication, transpose, inverse, and determinant) using the NumPy library.
4. Write a Python program to solve systems of linear equations and compute eigenvalues and eigenvectors using NumPy library.
5. Implement numerical methods for root finding and numerical integration using the SciPy library.
6. Write a Python program to perform interpolation and solve ordinary differential equations (ODEs) using SciPy library.
7. Write a Python program to load, clean, manipulate, and analyze datasets using the Pandas library.
8. Develop Python programs to create line plots, bar charts, scatter plots, histograms, and heatmaps using Matplotlib and Seaborn library.
9. Implement curve fitting, least-squares estimation, and optimization problems using the SciPy Optimize module.
10. Write a Python program to develop and evaluate regression and classification models using the Scikit-learn library.


36th BOS Date: 30-July-2026

Academic Batch: 2025 and Onwards

11. Implement a Python program to profile computational performance, compare execution time of algorithms, and optimize code using vectorization techniques.

[Handwritten signature]

Data Structures Lab			
Course Code:	CSE283	Course Credits:	2
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Labs (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Labs:	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Understand the implementation of fundamental data structures.			
2. Develop programs using linear and non-linear data structures.			
3. Apply appropriate data structures to solve computational problems.			
4. Analyze the performance of data structures and algorithms.			
5. Enhance programming, debugging, and problem-solving skills.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Implement array operations, recursion, and basic algorithm analysis.			
2. Develop and evaluate searching, sorting, and hashing algorithms.			
3. Implement linked lists, stacks, and queues for practical applications.			
4. Construct and manipulate tree and graph data structures using standard algorithms.			
5. Design efficient solutions by selecting appropriate data structures for real-world problems.			

List of Experiments:

1. Implement one-dimensional and two-dimensional array operations (insertion, deletion, searching, traversal, merging).
2. Implement recursive algorithms for Factorial and Fibonacci series. Analyze their time complexity.
3. Implement Linear Search and Binary Search. Compare their execution time and efficiency for different input sizes.
4. Implement Bubble Sort, Selection Sort, and Insertion Sort. Compare their time complexities experimentally.
5. Implement Merge Sort and Quick Sort. Compare recursive divide-and-conquer sorting algorithms.
6. Implement Heap Sort and Counting Sort algorithms.
7. Implement Hash Tables using Separate Chaining and Open Addressing.
8. Implement Singly and Circular Linked Lists with insertion, deletion, searching, and traversal operations.
9. Implement Stack using Arrays and Linked Lists.
10. Implement Circular Queue and Priority Queue using Arrays and Linked Lists.
11. Implement Binary Tree and Binary Search Tree operations including insertion, deletion, searching, and traversals (Preorder, Inorder, Postorder).
12. Implement Graph representations using Adjacency Matrix and Adjacency List. Perform BFS, DFS.

Handwritten signatures:

Database Management System Lab			
Course Code:	CSE285	Course Credits:	2
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	3U
No. of Labs (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Labs:	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To explain basic database concepts, applications, data models, schemas and instances.			
2. To demonstrate the use of constraints and relational algebra operations.			
3. To emphasize the importance of normalization in databases.			
4. To facilitate students in Database design			
5. To familiarize issues of concurrency control and transaction management.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Students get practical knowledge on designing and creating relational database systems.			
2. Understand various advanced queries execution such as relational constraints, joins, set operations, aggregate functions, trigger, views and embedded SQL			
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system			
4. Use the basics of SQL and construct queries using SQL in database creation and interaction.			
5. Analyze and Select storage and recovery techniques of database system.			

List of Experiments:

1. Introduction to MySQL, an exercise of data types in MySQL & Data Definition Language Commands.
2. Exercise on Data Manipulation Language and Transaction Control Commands.
3. Exercise of Type of Data Control Commands.
4. Exercise of joins (Single Table - Using Normalization)
5. Exercise of joins (Multiple Table - Using Normalization)
6. Exercise on GROUP BY/ORDER BY Clause and Date Arithmetic.
7. Exercise on different Functions (Aggregate, Math and String).
8. Exercise on different types of sub queries.
9. Exercise on Procedures.
10. Exercise on Views.

SEMESTER IV

Design and Analysis of Algorithms			
Course Code:	CSE202	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To analyze the asymptotic time and space performance of algorithms using appropriate complexity measures.			
2. To develop rigorous correctness proofs for algorithms using formal techniques such as invariants and induction.			
3. To acquire and demonstrate familiarity with fundamental and advanced algorithms and data structures.			
4. To apply key algorithmic design paradigms and analytical methods to solve computational problems effectively.			
5. To synthesize efficient and scalable algorithmic solutions for common engineering and real-world problem scenarios.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Analyze algorithms and data structures using asymptotic notations, recurrence relations, and amortized analysis to evaluate time and space complexity.			
2. Design efficient algorithms using appropriate paradigms such as greedy, divide-and-conquer, dynamic programming, backtracking, and branch and bound.			
3. Implement and apply advanced data structures and graph algorithms to solve complex computational problems.			
4. Design and evaluate advanced algorithmic solutions using randomized, approximation, and online paradigms with respect to correctness, efficiency, and practical applicability.			
5. Classify computational problems into complexity classes (P, NP, NP-Complete, NP-Hard) and justify algorithmic choices for intractable problems.			

UNIT I: INTRODUCTION

Introduction to Algorithm, Time Complexity and Space Complexity, Order of Growth, Asymptotic Notations: Big Oh, Theta, Omega, Amortized analysis, Analysing control statement, Loop Invariant, Recurrence Relations: Introduction, Substitution Method, Recursion Tree Method, Master's Theorem. Searching and Sorting Algorithms analysis.

UNIT II: ALGORITHM DESIGN TECHNIQUES

Divide and Conquer: Binary Search, Merge Sort, Strassen's Matrix Multiplication; Greedy Algorithm: General Characteristics, Knapsack Problem, Huffman code; Dynamic Programming: Introduction, Principle of Optimality, 0-1 Knapsack, Longest Common Subsequence, Matrix Chain Multiplication.

UNIT III: STATE SPACE SEARCH ALGORITHMS AND PATTERN MATCHING

Backtracking: State-Space Search Tree, N-Queen's problem, Graph Colouring; Branch and Bound: Travelling Salesman Problem; Pattern Matching: The Naïve string-matching algorithm, Rabin-Karp algorithm, Knuth-Morris-Pratt algorithm.

Handwritten signature

Handwritten signature

Handwritten mark

Handwritten mark

UNIT IV: ADVANCED GRAPH ALGORITHMS

Minimum Spanning Trees: Prim's algorithm, Kruskal's algorithm; Shortest Path: Bellman-Ford algorithm, Dijkstra's algorithm, Floyd Warshall algorithm; Network flow: Flow networks, Max-Flow, Min-Cut, Ford-Fulkerson method.

UNIT V: ADVANCED ALGORITHMS

P and NP, NP Complete and NP-Hard; Advanced Data Structures and their applications: Height Balanced Trees: AVL Trees, Balance Factor, Rotations, Insertion, Deletion, B-Trees, Insertion, Deletion, Advanced Algorithms and their applications.

Text Books:

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, C. Stein, Introduction to Algorithms, 4th Edition, MIT Press.
2. Anany Levitin, "Introduction to the Design and Analysis of Algorithm", Pearson Education Asia, 2003.
3. Sara Baase and Allen Van Gelder, "Computer Algorithms-Introduction to the Design and Analysis", Pearson Education Asia, 2003.
4. A. V. Aho, J.E. Hopcroft and J.D. Ullman, "The Design and Analysis of Computer Algorithms", Pearson Education Asia, 2003.

Computer Networks			
Course Code:	CSE204	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Describe how computer networks are organized with the concept of layered approach.			
2. Implement a simple LAN with hubs, bridges and switches.			
3. Analyze the contents in a given Data Link layer packet, based on the layer concept.			
4. Describe what classless addressing scheme is.			
5. Describe how routing protocols work.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Analyze the requirements for a given organizational structure and select the most appropriate networking architecture and technologies.			
2. Have a basic knowledge of the use of cryptography and network security.			
3. Specify and identify deficiencies in existing protocols, and then go onto formulate new and better protocols.			
4. Analyze specify and design the topological and routing strategies for an IP based networking infrastructure.			
5. Have a working knowledge of datagram and internet socket programming.			

UNIT I: INTRODUCTION AND PHYSICAL LAYER

Key concepts of computer network, transmission media, network devices, network topology, topology design issues, types of Network: LAN, MAN, WAN, PAN, ISDN systems and ATM network, OSI-reference model, open system standards, characteristics of network, TCP/IP model, protocols and standards, encoding technique.

UNIT II: SWITCHING AND DATA LINK LAYER

Multiplexing Techniques: TDMA, FDMA, WDMA, CDMA, Data link layer: LLC and MAC sublayers, protocols and design issues, issues IEEE 802 LAN Standards, framing, CRC, error control, flow control, HDLC, ALOHA and performance issues. Frames relay networks and performance parameters.

UNIT III: NETWORK LAYER

Network layer design issues, overview of IPv4 and IPv6, addressing: class full and classless, static and dynamic, subnet and super net, auto configuration through DHCP, routing protocols: RIP, DVR, LSR, OSPF, BGP, congestion control algorithm, subnet concept, virtual LAN, ICMP, multicasting, mobile IP.

UNIT IV: TRANSPORT LAYER

Port addressing schemes, connectionless and connection oriented services: TCP and UDP, wireless TCP, Congestion control, queue management, NAT, PAT, socket format at transport level, socket interface and programming.

Handwritten signatures and initials in blue ink.

UNIT V: APPLICATION LAYER

Client server architecture, domain name services, application services: HTTP, TELNET, RLOGIN, FTP, CBR, NFS, SMTP, POP, IMAP, MIME, voice and video over IP, social issues- privacy, freedom of speech, copy right.

Text Books:

1. S. Tanenbaum, Computer Networks, 4th edition, Prentice Hall, 2008.
2. Forouzan, B.A., Data Communication and Networking, Tata McGraw-Hill.
3. W. Stallings, Data and Computer Communications, 8th edition, Prentice Hall, 2007
4. Douglas E. Comer TCP/IP Principles, Protocols and Architecture, Pearson Education.

Handwritten signatures in blue ink:
A large signature on the left, a large signature in the middle, and a small signature on the right.

Artificial Intelligence			
Course Code:	CSE206	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Provide a strong foundation of fundamental concepts in Artificial Intelligence			
2. Enable the student to apply these techniques in applications which involve perception, reasoning and learning			
3. Provide a basic exposition to the goals and methods of Artificial Intelligence			
4. Explain the role of agents and how it is related to environment and the way of evaluating it and how agents can act by establishing goals.			
5. Learn the different machine learning techniques to design AI machine and enveloping applications for real world problems.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Understand the various searching techniques, constraint satisfaction problem and example problems- game playing techniques.			
2. Apply these techniques in applications which involve perception, reasoning and learning			
3. Acquire the knowledge of real-world Knowledge representation			
4. Analyze and design a real-world problem for implementation and understand the dynamic behavior of a system.			
5. To enable the student to apply these techniques in applications which involve perception, reasoning and learning			

UNIT I: Introduction

Introduction to Artificial Intelligence, History and Evolution of AI, Goals and Characteristics of AI, Types of AI (Narrow AI, General AI, Super AI), Applications of AI, Intelligent Systems, Turing Test, Rationality and Intelligence, AI Ethics and Responsible AI, Explainable AI, AI Safety and Limitations, Weak AI vs Strong AI, Overview of AI Paradigms: Symbolic AI, Statistical AI.

UNIT II: Intelligent Agents and Problem Solving

Intelligent Agents, Agent Architecture, Rational Agents, PEAS Representation, Environment Types, Types of Agents: Simple Reflex, Model-Based, Goal-Based, Utility-Based, Learning Agents, Agent Communication, Multi-Agent Systems (Introduction), Knowledge-Based Agents, Problem Formulation, State Space Representation.

UNIT III: Search Algorithms and Problem Solving

Problem Solving by Searching, State Space Search, Uninformed Search: Breadth First Search, Depth First Search, Uniform Cost Search; Informed Search: Best First Search, Greedy Best First Search, A* Search, Hill Climbing, Beam Search; Adversarial Search: Minimax Algorithm, Alpha-Beta Pruning; Introduction to Constraint Satisfaction Problems; Classical AI Problems and Applications: Route Finding, Puzzle Solving, Game Playing, Water-Jug Problem, Tic-Tac-Toe.

UNIT IV: Knowledge Representation and Reasoning

Knowledge Representation, Logic-Based Representation, Semantic Networks, Frames, Production Rules, Ontologies, Knowledge Engineering: Knowledge Acquisition, Representation, Storage, Knowledge Base Design, Knowledge-Based Systems, Reasoning: Forward Chaining, Backward Chaining, Rule-Based Reasoning, Uncertainty in AI, Introduction to Bayesian Reasoning, Fuzzy Logic.

UNIT V: AI Techniques and Applications

Introduction of Machine learning, Introduction of Deep Learning, Introduction to Expert system, Introduction to Natural Language Process, AI in future, AI in social media, AI in Entertainment and Education, AI in Drones, AI in Automated customer support, AI in Personalized shopping Experience, AI in Finance, AI in Smart cars, AI in Travel and Navigation, AI in Smart home Devices, AI in Security and Surveillance, AI in Education, AI in Health Care, AI in E-Commerce.

Text Books:

1. Artificial Intelligence, Russel and Norvig, Pearson Prentice Hall Publication, 2006.
2. Artificial Intelligence, Elanie Reich: Tata McGraw-Hill Publishing House, 2008.
3. Artificial Intelligence, Ela Kumar, IK Publishing.
4. Artificial intelligence, Peterson, Tata McGraw-Hill, 2008.
5. Artificial Intelligence, Winston, PHI publication, 2006.
6. Artificial Intelligence – A Modern Approach (3rd Edition) By Stuart Russell & Peter Norvig
7. Artificial Intelligence: The Basics By Kevin Warwick

Handwritten signature: H. K. Sankar

Handwritten signature: V. S. Sankar

Handwritten mark: P

Theory of Automata			
Course Code:	CSE208	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To introduce the fundamental concepts of automata, formal languages, and computability.			
2. To develop understanding of finite automata and regular expressions for pattern recognition and lexical analysis.			
3. To study context-free grammars and pushdown automata for syntax analysis and language processing.			
4. To understand the concepts of Turing machines, decidability, and computability.			
5. To provide theoretical foundations required for compiler design, artificial intelligence, and advanced computing systems.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Design deterministic and non-deterministic finite automata for regular languages.			
2. Apply regular expressions and finite automata for pattern matching and lexical analysis.			
3. Construct context-free grammars and pushdown automata for context-free languages.			
4. Analyze formal languages using pumping lemmas and closure properties.			
5. Design Turing machines and explain the concepts of decidability and computability.			

UNIT I: INTRODUCTION

Alphabets, Strings and Languages; Automata and Grammars, Deterministic finite Automata (DFA)- Formal Definition, Simplified notation: State transition graph, Transition table, Language of DFA, Nondeterministic finite Automata (NFA), NFA with epsilon transition, Language of NFA, Equivalence of NFA and DFA, Minimization of Finite Automata, Distinguishing one string from other, Myhill-Nerode Theorem.

UNIT II: REGULAR EXPRESSION (RE)

Definition, Operators of regular expression and their precedence, Algebraic laws for Regular expressions, Kleene's Theorem, Regular expression to FA, DFA to Regular expression, Arden Theorem, Non-Regular Languages, Pumping Lemma for regular Languages. Application of Pumping Lemma, Closure properties of Regular Languages, Decision properties of Regular Languages, FA with output: Moore and Mealy machine, Equivalence of Moore and Mealy Machine, Applications and Limitation of FA.

UNIT III: CONTEXT FREE GRAMMAR (CFG) AND CONTEXT FREE LANGUAGES (CFL)

Context free grammar (CFG) and Context Free Languages (CFL): Definition, Examples, Derivation, Derivation trees, Ambiguity in Grammar, Inherent ambiguity, Ambiguous to Unambiguous CFG, Useless symbols, Simplification of CFGs, Normal forms for CFGs: CNF and GNF, Closure properties of CFLs, Decision Properties of CFLs: Emptiness, Finiteness and Membership, Pumping lemma for CFLs.

UNIT IV: PUSH DOWN AUTOMATA (PDA)

Description and definition, Instantaneous Description, Language of PDA, Acceptance by Final state, Acceptance by empty stack, Deterministic PDA, Equivalence of PDA and CFG, CFG to PDA and PDA to CFG, two stack PDA.

Helanka

36th BoS Date: 30-July-2026

Academic Batch: 2025 and Onwards

UNIT V: TURING MACHINES (TM)

Basic model, definition and representation, Instantaneous Description, Language acceptance by TM, Variants of Turing Machine, TM as Computer of Integer functions, Universal TM, Church's Thesis, Recursive and recursively enumerable languages, Halting problem.

Text/Reference Books:

1. Hopcroft, Ullman, "Introduction to Automata Theory, Languages and Computation", Pearson Education.
2. Peter Linz, "An Introduction to Formal Language and Automata", Narosa Publishing house.
3. YN Singh "Mathematical Foundation of Computer Science", New Age International.
4. Malviya, AK "Theory of Computation and Application", BPaperback Publications
5. Papadimitrou, C. and Lewis, CL, "Elements of the Theory of Computation", Pearson Publication.
6. K. Krithivasan and R. Rama; Introduction to Formal Languages, Automata Theory and Computation; Pearson Education.
7. KLP Mishra and N. Chandrasekaran, "Theory of Computer Science: Automata, Languages and Computation", PHI Learning Private Limited, Delhi India.
8. Harry R. Lewis and Christos H. Papadimitriou, Elements of the theory of Computation, Second Edition, Prentice-Hall of India Pvt. Ltd.
9. Micheal Sipser, "Introduction of the Theory and Computation", Thomson Learning.
10. Katuri Viswanath, "Introduction to Mathematical Computer Science, An" Universities Press.

[Handwritten signatures in blue ink]

Design and Analysis of Algorithms Lab			
Course Code:	CSE282	Course Credits:	2
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Lectures (L + T):	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Implement and experimentally analyze algorithms using appropriate measures of time and space complexity.			
2. Develop solutions using divide-and-conquer, greedy, dynamic programming, and backtracking paradigms.			
3. Implement tree, heap, and graph structures required for efficient algorithmic problem solving.			
4. Evaluate alternative algorithms on varied input characteristics and justify an appropriate choice.			
5. Develop correct, readable, and testable programs using C, Java, or Python.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Implement algorithms and empirically analyze their best-, average-, and worst-case behaviour.			
2. Design and implement divide-and-conquer and greedy algorithms for sorting, selection, spanning tree, shortest path, scheduling, coding, and knapsack problems.			
3. Formulate and implement dynamic programming and backtracking solutions and reconstruct the corresponding optimal or feasible solution.			
4. Implement and apply heaps, search trees, balanced trees, and graph traversal techniques.			
5. Implement standard string-matching algorithms and compare competing algorithms using reproducible experiments.			

PRACTICALS

Note: Students may use C, Java, or Python. Each experiment must include algorithm/pseudocode, source code, test cases, output, and a brief complexity/performance analysis.

Lab 1: Experimental Analysis of Algorithms

1. Implement iterative and recursive versions of factorial and Fibonacci computation; count basic operations and compare their running times for increasing input sizes.
2. Implement linear search and binary search; test best-, average-, and worst-case inputs and plot input size versus execution time.

Lab 2: Elementary Sorting and Heap-Based Processing

1. Implement bubble sort, insertion sort and selection sort; record comparisons and data movements for sorted, reverse-sorted, and random inputs.

Lab 3: Divide-and-Conquer Fundamentals

1. Implement binary search using both iterative and recursive approaches; compare their execution time and auxiliary space.
2. Implement merge sort; test it on arrays containing duplicate, negative, sorted, reverse-sorted, and random values, and analyze its time and auxiliary space.

Lab 4: Advanced Divide-and-Conquer Algorithms

1. Implement quick sort using first-element, last-element, randomized, and median-of-three pivot selection; compare performance on different input distributions.
2. Implement Strassen's matrix multiplication and compare its correctness and execution time with conventional matrix multiplication for comparable matrix sizes.

Handwritten signature: Anand

36th BoS Date: 30-July-2026

Academic Batch: 2025 and Onwards

Lab 5: Greedy Algorithms for Spanning Trees and Scheduling

1. Implement Prim's algorithm to obtain a minimum spanning tree of a weighted, connected, undirected graph and report its total cost.
2. Implement Kruskal's algorithm using disjoint-set operations; compare its output and running time with Prim's algorithm on sparse and dense graphs.

Lab 6: Greedy Shortest Paths, Coding, and Knapsack

1. Implement Dijkstra's algorithm for a graph with non-negative edge weights; display the shortest distance and path from a selected source to every vertex.
2. Implement the fractional knapsack problem; display the selected fraction of each item and compare the result with a simple value-based selection strategy.

Lab 7: Dynamic Programming - Paths and Knapsack

1. Implement the 0/1 knapsack problem using memoization and tabulation; reconstruct the selected items and compare the two implementations.
2. Implement the Floyd-Warshall algorithm for all-pairs shortest paths; construct the distance matrix and identify whether a negative cycle exists.

Lab 8: Dynamic Programming and Backtracking

1. Implement the traveling salesperson problem using dynamic programming for small instances; report the minimum tour cost and the tour.
2. Implement the N-Queens problem using backtracking; generate all valid solutions for a given board size and report the number of states explored.

Lab 9: String Matching and Comparative Study

1. Implement the Naive and Rabin-Karp string-matching algorithms; report all match positions, spurious hits, and character comparisons.
2. Implement the Knuth-Morris-Pratt algorithm; construct and display the prefix table and verify matches for patterns with repeated prefixes.

Lab 10: Advanced Trees and Graph Traversal

1. Implement a binary search tree with insertion, deletion, search, and inorder, preorder, and postorder traversals; verify the BST property after updates.
2. Implement an AVL tree with all required rotations; compare its height and search cost with an ordinary BST for the same insertion sequence.

Computer Networks Lab			
Course Code:	CSE284	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Lectures (L + T):	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Practical knowledge of working principles of various communication protocols.			
2. Analyze structure and formats of TCP/IP layer protocols.			
3. Understanding of networking fundamentals.			
4. Understanding of learning the process of Internet of Things applications planning.			
5. Understanding of configuration of various end devices, server, routers and switches.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Understand the practical approach to network communication protocols.			
2. Understand network layers, structure/format and role of each network layer.			
3. Able to design and implement various network application such as data transmission between client and server, file transfer, real-time multimedia transmission.			
4. Understand the various Routing Protocols/Algorithms and Internetworking.			
5. Learn to configure server.			

List of Experiments:

1. Introduction to transmission media (CAT5, OFC, COAXIAL CABLE Wireless).
2. Introduction to network interfaces (Wired and Wireless).
3. Configure and installing an Ethernet (10/100).
4. Performance evaluation of Ethernet (10/100).
5. Topology design (Ring, Bus).
6. Generation of data packet and measurement (CBR, VBR, Poison).
7. Implement the following:
 - a. Router Configuration
 - b. Server Configuration
 - c. Server Configuration
8. Congestion control of network and QoS of network
9. Protocols and the configuration
10. Security (WEP, WPA) and Qualnet.





B.Tech.
CSE
Specialization Core Courses
SEMESTER IV



Machine Learning			
Course Code:	CSE210	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To understand the fundamental concepts, paradigms, and applications of Machine Learning.			
2. To apply supervised and unsupervised learning techniques to solve real-world problems.			
3. To understand the principles of artificial neural networks and deep learning models.			
4. To explore reinforcement learning and recent advancements in machine learning techniques.			
5. To analyze and select appropriate machine learning algorithms for data-driven decision-making.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Explain the concepts, types, and applications of machine learning.			
2. Implement and evaluate supervised learning algorithms for regression and classification tasks.			
3. Develop artificial neural network models using appropriate activation functions and learning algorithms.			
4. Apply unsupervised learning techniques for clustering and dimensionality reduction.			
5. Analyze reinforcement learning, deep learning models, and emerging machine learning techniques for solving practical problems.			

UNIT I INTRODUCTION TO MACHINE LEARNING

What is Machine Learning, Types of Machine Learning, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Applications of Machine Learning – Stock Price Prediction, Face Recognition, Handwriting Recognition, Image Recognition, Virtual Personal Assistants, Medical Diagnosis, Online Fraud Detection.

UNIT II SUPERVISED LEARNING (REGRESSION/CLASSIFICATION)

Linear models: Linear Regression, Logistic Regression, Generalized Linear Models; Basic methods: Distance-based methods, Nearest-Neighbours, Decision Trees, Naive Bayes, Support Vector Machines, Nonlinearity and Kernel Methods, Ensemble Methods (Boosting, Bagging, Random Forests).

UNIT III ARTIFICIAL NEURAL NETWORKS

Biological Neurons and Biological Neural Networks, Artificial Neural Network, Types of Neural Network, Perceptron, History behind Perceptron, Importance of Perceptron, Working of Perceptron, Perceptron Learning, Perceptron Learning Rule, Perceptron Learning of AND & OR gate, XOR gate, Activation functions, Binary Activation function, ReLU, Sigmoid, Hyperbolic, Softmax Activation function, Multilayer Perceptrons, Back propagation Neural Networks, and Feed-Forward Neural Networks, Applications and Future of Neural Networks.

UNIT IV UNSUPERVISED LEARNING

Clustering: K-means/Kernel K-means, Self-Organizing Maps. Dimensionality Reduction: PCA and kernel PCA, Matrix Factorization and Matrix Completion Generative Models (mixture models and latent factor models).

Apalank
36th BoS Date: 30-July-2026

Academic Batch: 2025 and Onwards

UNIT V REINFORCEMENT LEARNING AND DEEP LEARNING

Introduction to Reinforcement Learning, Learning Task, Example of Reinforcement Learning in Practice. Deep Learning: Sparse Modeling and Estimation, Modeling Sequence/Time-Series Data, Deep Learning and Feature Representation Learning. Recent trends in various learning techniques of machine learning and classification methods, Case studies in interdisciplinary domains.

Text/ Reference Books:

1. Tom M. Mitchell, Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
3. Stephen Marsland, Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.
5. Hastie, T., Tibshirani, R., Friedman, J., The Elements of Statistical Learning, Springer, 2008.
6. James, G., Witten, D., Hastie, T., Tibshirani, R., An Introduction to Statistical Learning, Springer, 2013.

Handwritten signatures in blue ink:
Aulanki, Vaylath, and a small signature.

Computer Graphics and Multimedia			
Course Code:	CSE212	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Understand the fundamentals of computer graphics and multimedia systems.			
2. Apply geometric transformations and viewing techniques to graphical objects.			
3. Analyze algorithms for graphics rendering, clipping, and visualization.			
4. Explore multimedia components, compression, and animation techniques.			
5. Develop basic graphics and multimedia applications using standard tools and APIs.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Explain the concepts of computer graphics, display systems, and multimedia.			
2. Apply 2D/3D transformations, viewing, and clipping techniques to graphical objects.			
3. Analyze rendering, illumination, and hidden surface removal algorithms.			
4. Demonstrate multimedia concepts, compression techniques, and animation principles.			
5. Develop basic graphics and multimedia applications using modern graphics tools and APIs.			

UNIT I: INTRODUCTION TO COMPUTER GRAPHICS

Introduction to Computer Graphics and Multimedia, Applications of Computer Graphics, Graphics Hardware and Software, Graphics Systems and Display Devices, Raster Scan and Random Scan Displays, Input and Output Devices, Frame Buffer, Graphics Primitives, Line Drawing Algorithms (DDA and Bresenham), Circle Drawing Algorithms (Midpoint Circle Algorithm).

UNIT II: TWO-DIMENSIONAL GRAPHICS

Coordinate Systems, Windowing and Viewport Transformation, Two-Dimensional Transformations, Composite Transformations, Homogeneous Coordinates, Line Clipping (Cohen-Sutherland, Liang-Barsky), Polygon Clipping (Sutherland-Hodgman), Polygon Filling Algorithms, Introduction to Curves (Bezier and B-Spline).

UNIT III: THREE-DIMENSIONAL GRAPHICS

Three-Dimensional Graphics Concepts, 3D Coordinate Systems, 3D Transformations, Projections (Orthographic and Perspective), Visible Surface Detection, Back Face Detection, Hidden Surface Removal (Z-buffer Algorithm), Illumination Models, Introduction to Shading (Flat, Gouraud, Phong).

UNIT IV: MULTIMEDIA SYSTEMS

Introduction to Multimedia, Multimedia Components (Text, Images, Audio, Video, Animation), Multimedia Hardware and Software, Image Representation and Color Models (RGB, CMYK, HSV), Audio Fundamentals, Video Fundamentals, Image and Video

Handwritten signature

36th BoS Date: 30-July-2026

Handwritten signature

Handwritten signature

Academic Batch: 2025 and Onwards

UNIT V: COMPUTER ANIMATION AND EMERGING GRAPHICS

Principles of Animation, Animation Techniques (Keyframe, Tweening), 2D and 3D Animation, Introduction to OpenGL Graphics Pipeline, Texture Mapping, Introduction to Virtual Reality and Augmented Reality, Scientific Visualization, Game Graphics, Multimedia in Web and Mobile Applications, Emerging Trends in Computer Graphics.

Text / Reference Books:

1. Donald Hearn and M. Pauline Baker, Computer Graphics with OpenGL, 4th Edition, Pearson.
2. Foley, van Dam, Feiner, and Hughes, Computer Graphics: Principles and Practice, 3rd Edition, Pearson.
3. Ralf Steinmetz and Klara Nahrstedt, Multimedia: Computing, Communications and Applications, Pearson.
4. Ze-Nian Li, Mark S. Drew, and Jiangchuan Liu, Fundamentals of Multimedia, 3rd Edition, Springer.
5. John F. Hughes et al., Computer Graphics: Principles and Practice, Addison-Wesley.

Machine Learning Lab			
Course Code:	CSE286	Course Credits:	2
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Lectures (L + T):	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Develop proficiency in Python programming and machine learning libraries for data analysis.			
2. Implement and evaluate supervised and unsupervised machine learning algorithms.			
3. Apply appropriate data pre-processing, visualization, and model evaluation techniques.			
4. Perform model optimization through hyperparameter tuning and comparative analysis.			
5. Solve real-world machine learning problems using standard tools and datasets.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Use Python libraries to perform data pre-processing, statistical analysis, and visualization.			
2. Implement and evaluate regression and classification models using appropriate performance metrics.			
3. Apply hyperparameter tuning techniques to improve the performance of machine learning models.			
4. Implement clustering techniques for document and data analysis using suitable evaluation measures.			
5. Develop complete machine learning solutions for real-world datasets using standard Python-based frameworks.			

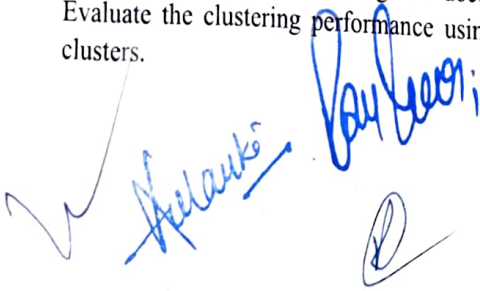
List of Experiments:

1. Implement a Python program to compute measures of central tendency (Mean, Median, and Mode) and measures of dispersion (Variance and Standard Deviation).
2. Implement basic functions of Python libraries such as statistics, math, NumPy, and SciPy for scientific computing.
3. Implement basic functions of Python libraries for machine learning applications, including Pandas and Matplotlib, for data manipulation and visualization.
4. Implement Linear Regression using five datasets of varying sizes. Compare the model performance by plotting evaluation metrics such as MAE, MSE, RMSE, and R^2 Score.
5. Implement the K-Nearest Neighbors (KNN) classifier using scikit-learn. Evaluate the model on five datasets of varying sizes by comparing Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Using one large dataset, vary the value of K (1, 3, 5, 7, 9, 11, 21, and 31) and compare the performance using the same evaluation metrics.
6. Implement the Decision Tree classifier using scikit-learn and perform hyperparameter tuning. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.






7. Implement the Naïve Bayes classifier using scikit-learn. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
8. Implement the Support Vector Machine (SVM) classifier using scikit-learn. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
9. Implement the Random Forest classifier using scikit-learn. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
10. Implement ensemble learning algorithms (AdaBoost and XGBoost). Evaluate the models on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
11. Implement an Artificial Neural Network (ANN) using scikit-learn or TensorFlow/Keras and perform hyperparameter tuning. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
12. Implement K-Means clustering for document clustering using suitable text datasets. Evaluate the clustering performance using Silhouette Score and visualize the resulting clusters.

Three handwritten signatures in blue ink are present below the list. The first signature on the left is partially cut off. The middle signature is written in a cursive style. The signature on the right is also cursive and appears to be 'Vandana'.

B.Tech.
Data Science
Specialization Core Courses
SEMESTER IV




Machine Learning			
Course Code:	DSE210	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To understand the fundamental concepts, paradigms, and applications of Machine Learning.			
2. To apply supervised and unsupervised learning techniques to solve real-world problems.			
3. To understand the principles of artificial neural networks and deep learning models.			
4. To explore reinforcement learning and recent advancements in machine learning techniques.			
5. To analyze and select appropriate machine learning algorithms for data-driven decision-making.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Explain the concepts, types, and applications of machine learning.			
2. Implement and evaluate supervised learning algorithms for regression and classification tasks.			
3. Develop artificial neural network models using appropriate activation functions and learning algorithms.			
4. Apply unsupervised learning techniques for clustering and dimensionality reduction.			
5. Analyze reinforcement learning, deep learning models, and emerging machine learning techniques for solving practical problems.			

UNIT I INTRODUCTION TO MACHINE LEARNING

What is Machine Learning, Types of Machine Learning, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Applications of Machine Learning – Stock Price Prediction, Face Recognition, Handwriting Recognition, Image Recognition, Virtual Personal Assistants, Medical Diagnosis, Online Fraud Detection.

UNIT II SUPERVISED LEARNING (REGRESSION/CLASSIFICATION)

Linear models: Linear Regression, Logistic Regression, Generalized Linear Models; Basic methods: Distance-based methods, Nearest-Neighbours, Decision Trees, Naive Bayes, Support Vector Machines, Nonlinearity and Kernel Methods, Ensemble Methods (Boosting, Bagging, Random Forests).

UNIT III ARTIFICIAL NEURAL NETWORKS

Biological Neurons and Biological Neural Networks, Artificial Neural Network, Types of Neural Network, Perceptron, History behind Perceptron, Importance of Perceptron, Working of Perceptron, Perceptron Learning, Perceptron Learning Rule, Perceptron Learning of AND & OR gate, XOR gate, Activation functions, Binary Activation function, ReLU, Sigmoid, Hyperbolic, Softmax Activation function, Multilayer Perceptrons, Back propagation Neural Networks, and Feed-Forward Neural Networks, Applications and Future of Neural Networks.

UNIT IV UNSUPERVISED LEARNING

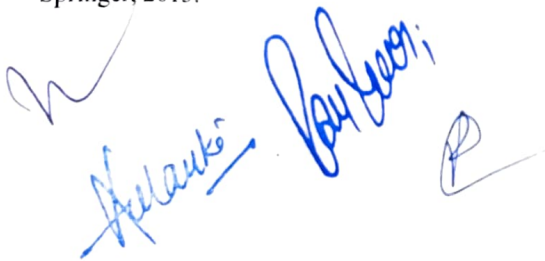
Clustering: K-means/Kernel K-means, Self-Organizing Maps. Dimensionality Reduction: PCA and kernel PCA, Matrix Factorization and Matrix Completion Generative Models (mixture models and latent factor models).

UNIT V REINFORCEMENT LEARNING AND DEEP LEARNING

Introduction to Reinforcement Learning, Learning Task, Example of Reinforcement Learning in Practice. Deep Learning: Sparse Modeling and Estimation, Modeling Sequence/Time-Series Data, Deep Learning and Feature Representation Learning. Recent trends in various learning techniques of machine learning and classification methods, Case studies in interdisciplinary domains.

Text/ Reference Books:

1. Tom M. Mitchell, Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
3. Stephen Marsland, Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.
5. Hastie, T., Tibshirani, R., Friedman, J., The Elements of Statistical Learning, Springer, 2008.
6. James, G., Witten, D., Hastie, T., Tibshirani, R., An Introduction to Statistical Learning, Springer, 2013.

Three handwritten signatures in blue ink are present below the list of books. The first signature is on the left, the second is in the middle and appears to be 'Hastie', and the third is on the right.

Data Mining and Warehousing			
Course Code:	DSE212	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Understand the fundamentals of data warehousing and data mining.			
2. Apply data preprocessing and transformation techniques for knowledge discovery.			
3. Discover patterns, associations, and correlations from large datasets.			
4. Perform clustering and outlier analysis using data mining algorithms.			
5. Explore advanced data mining techniques and real-world applications.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Explain the concepts of data warehousing and knowledge discovery.			
2. Apply preprocessing techniques to prepare data for mining.			
3. Discover frequent patterns and association rules from datasets.			
4. Perform clustering and outlier detection for data analysis.			
5. Analyze complex data using advanced data mining techniques and tools.			

UNIT I: DATA WAREHOUSING AND KNOWLEDGE DISCOVERY

Introduction to Data Mining and Knowledge Discovery (KDD), Data Mining Functionalities and Applications, Data Warehouse Architecture, OLTP vs OLAP, ETL Process, Data Marts, Metadata, Dimensional Modeling, Fact Tables and Dimension Tables, Star Schema and Snowflake Schema, Data Cubes, OLAP Operations: Roll-up, Drill-down, Slice, Dice, Pivot.

UNIT II: DATA PREPROCESSING AND DATA EXPLORATION

Types of Data, Data Quality Issues, Data Cleaning, Data Integration, Data Transformation, Data Reduction, Data Discretization, Concept Hierarchies, Feature Extraction, Feature Selection, Similarity and Dissimilarity Measures, Data Visualization for Mining, Introduction to Dimensionality Reduction (PCA, CUR).

UNIT III: PATTERN DISCOVERY AND ASSOCIATION ANALYSIS

Concept Description, Data Characterization, Data Discrimination, Attribute Relevance Analysis, Frequent Pattern Mining, Apriori Algorithm, FP-Growth Algorithm, Closed and Maximal Frequent Itemsets, Association Rule Mining, Correlation Analysis, Interestingness Measures, Constraint-Based Association Mining, Sequential Pattern Mining, Market Basket Analysis.

UNIT IV: CLUSTERING AND OUTLIER ANALYSIS

Clustering Fundamentals, Similarity Measures, Partitioning Methods (K-Means, K-Medoids/PAM), Hierarchical Clustering, Density-Based Clustering (DBSCAN), Cluster

Evaluation, Outlier Detection, High-Dimensional Data Clustering, Graph and Network Clustering, Spatial Clustering.

UNIT V: ADVANCED DATA MINING

Text Mining, Web Mining, Web Usage Mining, Graph Mining, Social Network Mining, Stream Data Mining, Temporal Data Mining, Spatial Data Mining, Multimedia Data Mining, Privacy-Preserving Data Mining, Data Mining Applications in Healthcare, Finance, E-commerce, Cybersecurity, and Smart Cities, Introduction to WEKA and RapidMiner/Python-based Data Mining.

Text/ Reference Books:

1. Jiawei Han, Jian Pei, Hanghang Tong, Data Mining: Concepts and Techniques, 4th Edition, Morgan Kaufmann, 2022.
2. Margaret H. Dunham, Data Mining: Introductory and Advanced Topics, Pearson Education.
3. Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar, Introduction to Data Mining, 2nd Edition, Pearson.
4. Alex Berson and Stephen J. Smith, "Data Warehousing, Data Mining & OLAP", Tata McGraw Hill Edition, Tenth Reprint 2007.
5. Ralph Kimball and Margy Ross, The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling, 3rd Edition, Wiley.
6. Ian H. Witten, Eibe Frank, Mark A. Hall, Christopher J. Pal, Data Mining: Practical Machine Learning Tools and Techniques, 4th Edition, Morgan Kaufmann.
7. M. Kantardzic, "Data mining: Concepts, models, methods and algorithms, John Wiley & Sons Inc.
8. M. Dunham, "Data Mining: Introductory and Advanced Topics", Pearson Education.

Machine Learning Lab			
Course Code:	DSE286	Course Credits:	2
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Lectures (L + T):	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. Develop proficiency in Python programming and machine learning libraries for data analysis.			
2. Implement and evaluate supervised and unsupervised machine learning algorithms.			
3. Apply appropriate data pre-processing, visualization, and model evaluation techniques.			
4. Perform model optimization through hyperparameter tuning and comparative analysis.			
5. Solve real-world machine learning problems using standard tools and datasets.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Use Python libraries to perform data pre-processing, statistical analysis, and visualization.			
2. Implement and evaluate regression and classification models using appropriate performance metrics.			
3. Apply hyperparameter tuning techniques to improve the performance of machine learning models.			
4. Implement clustering techniques for document and data analysis using suitable evaluation measures.			
5. Develop complete machine learning solutions for real-world datasets using standard Python-based frameworks.			

List of Experiments:

1. Implement a Python program to compute measures of central tendency (Mean, Median, and Mode) and measures of dispersion (Variance and Standard Deviation).
2. Implement basic functions of Python libraries such as statistics, math, NumPy, and SciPy for scientific computing.
3. Implement basic functions of Python libraries for machine learning applications, including Pandas and Matplotlib, for data manipulation and visualization.
4. Implement Linear Regression using five datasets of varying sizes. Compare the model performance by plotting evaluation metrics such as MAE, MSE, RMSE, and R² Score.
5. Implement the K-Nearest Neighbors (KNN) classifier using scikit-learn. Evaluate the model on five datasets of varying sizes by comparing Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Using one large dataset, vary the value of K (1, 3, 5, 7, 9, 11, 21, and 31) and compare the performance using the same evaluation metrics.
6. Implement the Decision Tree classifier using scikit-learn and perform hyperparameter tuning. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

[Handwritten signatures]

7. Implement the Naïve Bayes classifier using scikit-learn. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
8. Implement the Support Vector Machine (SVM) classifier using scikit-learn. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
9. Implement the Random Forest classifier using scikit-learn. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
10. Implement ensemble learning algorithms (AdaBoost and XGBoost). Evaluate the models on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
11. Implement an Artificial Neural Network (ANN) using scikit-learn or TensorFlow/Keras and perform hyperparameter tuning. Evaluate the model on five datasets of varying sizes and compare Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
12. Implement K-Means clustering for document clustering using suitable text datasets. Evaluate the clustering performance using Silhouette Score and visualize the resulting clusters.



B.Tech.

Cyber Security

Specialization Core Courses

SEMESTER IV



Cryptography and Network Security			
Course Code:	CSY210	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To understand the mathematics behind Cryptography.			
2. To understand the security concerns and vulnerabilities.			
3. To familiarize with different types of cryptosystems.			
4. To create an awareness for the design of various cryptographic primitives.			
5. To analyze different types of attacks on various cryptosystems.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. To learn the importance of number theory in designing crypto systems.			
2. To design public and private key cryptosystems.			
3. To do cryptanalysis of various cryptosystems.			
4. To implement cryptographic algorithms.			
5. To structure Privacy issues and able to resolve them.			

UNIT I: SECURITY FUNDAMENTALS AND CRYPTOGRAPHY

Introduction to Security-Security Goals – Security services(Confidentiality, Integrity, Authentication, Non-repudiation, Access control) – Security Mechanisms (Encipherment, Data Integrity, Digital Signature, Authentication Exchange, Traffic Padding, Routing Control, Notarization, Access control) - Security Principles. Introduction to Cryptography:- Kerckhoff's Principle -Classification of Cryptosystems-Cryptanalytic attacks- Cipher Properties (Confusion, Diffusion).

UNIT II: CLASSICAL AND MODERN SYMMETRIC CIPHERS

Traditional Secret Key Ciphers: Substitution Ciphers (mono alphabetic ciphers, poly alphabetic ciphers)-Transposition Ciphers-Stream and Block Ciphers. Modern Secret Key Ciphers:- Substitution Box-Permutation Box-Product Ciphers

UNIT III: BLOCK CIPHERS AND HASH FUNCTIONS

Data Encryption Standard (DES) (Fiestel and Non-Fiestel Ciphers, Structure of DES, DES Attacks, 2-DES, 3-DES) - Advanced Encryption Standard (AES) (Structure, Analysis)- Cryptographic Hash Functions– Properties - Secure Hash Algorithm-Message Authentication Code (MAC).

UNIT IV: PUBLIC KEY CRYPTOGRAPHY

Public Key Cryptosystems (PKC): - Types of PKC –Trapdoor -one way functions -RSA Cryptosystem (Integer Factorisation Trapdoor, Key Generation, Encryption, Decryption) - El Gamal Cryptosystem (Discrete Logarithm Trapdoor, Key Generation, Encryption, Decryption) - Diffie-Hellman Key Exchange Protocol, Man in the Middle attack on Diffie-Hellman Protocol.

UNIT V: DIGITAL SIGNATURES AND NETWORK SECURITY

Digital Signature: Signing – Verification - Digital signature forgery (Existential forgery, Selective forgery, Universal forgery) - RSA Digital Signature Scheme - ElGamal Signature Scheme - IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload- Intruders, Intrusion Detection, Distributed Denial of Service attacks

Text /Reference Books:

1. Behrouz A. Forouzan and Debdeep Mukhopadhyay, Cryptography & Network Security, Second Edition, Tata McGraw Hill, New Delhi, 2010
2. Douglas R. Stinson, "Cryptography: Theory and Practice", Third Edition, CRC Press.
3. William Stallings, "Cryptography and Network Security – Principles and Practices", Pearson Education, Fourth Edition, 2006.
4. Atul Kahate, "Cryptography and Network Security", 2nd Edition, Tata McGraw Hill, 2003.
5. Bernard Menezes, Network Security and Cryptography-Cengage Learning India, 2011.
6. Bruce Schneier, "Applied Cryptography: Protocols, Algorithms, and Source Code in C", Second Edition, John Wiley and Sons Inc, 2001.
7. Thomas Mowbray, "Cybersecurity Managing Systems Conducting Testing, and Investigating Intrusions", John Wiley, 2013.
8. Wenbo Mao, "Modern Cryptography: Theory & Practice", Pearson Education, 2006.

Introduction to Cyber Security			
Course Code:	CSY212	Course Credits:	3
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	03 + 00	Mid Sem. Exam Hours:	1.5
Total No. of Lectures (L + T):	45	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To prepare students with the technical knowledge and skills needed to protect and defend computer systems and networks.			
2. To develop graduates that can plan, implement, and monitor cyber security mechanisms to help ensure the protection of information technology assets.			
3. Analyze and resolve security issues in networks and computer systems to secure an IT infrastructure.			
4. Evaluate and communicate the human role in security systems with an emphasis on ethics, social engineering vulnerabilities and training.			
5. Interpret and forensically investigate security incident.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Follow a structured model in Security Systems Development Life Cycle (SDLC)			
2. Plan, implement and audit operating systems' security in a networked, multi-platform and cross platform environment			
3. Protect data and respond to threats that occur over the Internet			
4. Design and implement risk analysis, security policies, and damage assessment			
5. Detect attack methodology and combat hackers from intrusion or other suspicious attempts at connection to gain unauthorized access to a computer and its resources			

UNIT I: CYBER SECURITY FUNDAMENTALS

Overview of cyber security and why it is important; Internet governance: challenges and constraints; cyber threats: cyber warfare, cybercrime, cyber terrorism, cyber espionage; Need for a comprehensive cyber security policy; Need for an international convention on cyber space; Identify trends in cyber security events and protection techniques;

UNIT II: CYBER SECURITY VULNERABILITIES

Categorize assets, risk, threat and vulnerability; Identify different types of vulnerabilities in a system; Vulnerability in software; Complex network architecture of organization; Open access to organizational data; Weak authentication; Determine the phase of a cyber-attack; Overview of cryptography; Deception; Denial of service filters; Ethical hacking

UNIT III: CYBERSPACE AND THE LAW

Computer Ethics and security policies; Cyber security regulations; Role of International Laws; Role of stakeholders (state and private sector) in cyber security; Components of cyber security framework; Cyber security standards; Indian cyber space; National cyber security policy;

UNIT IV: CYBER FORENSICS

Overview- Why cyber forensics is important; Types of computer forensics; Objective of cyber security forensics investigators; How experts works; Stages of forensics investigation; Techniques and tools used by forensics experts; Advantages of cyber forensics; Incident handling;

UNIT V: SECURING WEB APPLICATIONS, SERVICES AND SERVER

Introduction; Basic security for HTTP applications and services; Basic security for SOAP (Simple Object Access Protocol)services; Identify management and Web Services; Authorization Patterns; Security Considerations and challenges;

Text Books:

1. Jon Friedman. Mark Bouchard, CISSP. Foreword by John P. Watters to. Cyber Threat Intelligence. Definitive GuideTM. 2015.
2. Scott Roberts, Rebekah Brown. Intelligence-Driven Incident Response: Outwitting the Adversary: O'Reilly 2017.
3. Bob Gourley, The Cyber Threat, Createspace Independent Pub 2014
4. Marjie T Britz, Computer Forensics and Cyber Crime: An Introduction, Pearson Education, 2nd Edition, 2008.
5. Cyberspace and the Law: Your Rights and Duties in the On-Line World By Edward Cavazos and Gavino Morin MIT Press: 1994.

Cryptography and Network Security Lab			
Course Code:	CSY286	Course Credits:	2
Course Category:	PCC	Course (U / P)	U
Course Year (U / P):	2U	Course Semester (U / P):	4U
No. of Lectures + Tutorials (Hrs/Week):	1 (4 HRS)	Mid Sem. Exam Hours:	-
Total No. of Lectures (L + T):	10	End Sem. Exam Hours:	3
COURSE OBJECTIVES			
1. To learn different commands for implementing encryption and decryption.			
2. To understand the implementation of Caesar cipher.			
3. To understand the encryption and decryption for XOR operation.			
4. To understand the working of wireless audit on a router.			
5. To demonstrate intrusion detection system.			
COURSE OUTCOMES			
At the end of the course the students should be able to:			
1. Apply encryption and decryption on any input.			
2. Generate random numbers within a range.			
3. Apply an RSA algorithm with different values of p and q.			
4. Apply snort tool.			
5. Design New policies related to privacy methods			

List of Experiments:

1. To understand and illustrate the different commands used to implement encryption and decryption.
2. Write an encryption and decryption program for Caesar cipher.
3. Write a program in which encryption and decryption is done in one program without input.
4. Write a program in C++ to generate Pseudo Random numbers in a range.
5. Write a program in C++ for XOR Encryption and Decryption.
6. Write a program in C++ for Vernam Cipher.
7. Write a program in C++ for the RSA algorithm taking p and q randomly.
8. Working with KfSensor Tool for Creating And Monitoring Honeypot.
9. Working with NetStumbler to Perform Wireless Audit On Router.
10. Working with Snort Tool to Demonstrate Intrusion Detection System

Arulank

Barbar

[Signature]

[Signature]